

Chapter 1

One shape

Draw a shape. Then change one thing about it, and see what happens.

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One circle

This is a whole sketch. Run it, and a circle appears in the middle of the canvas.

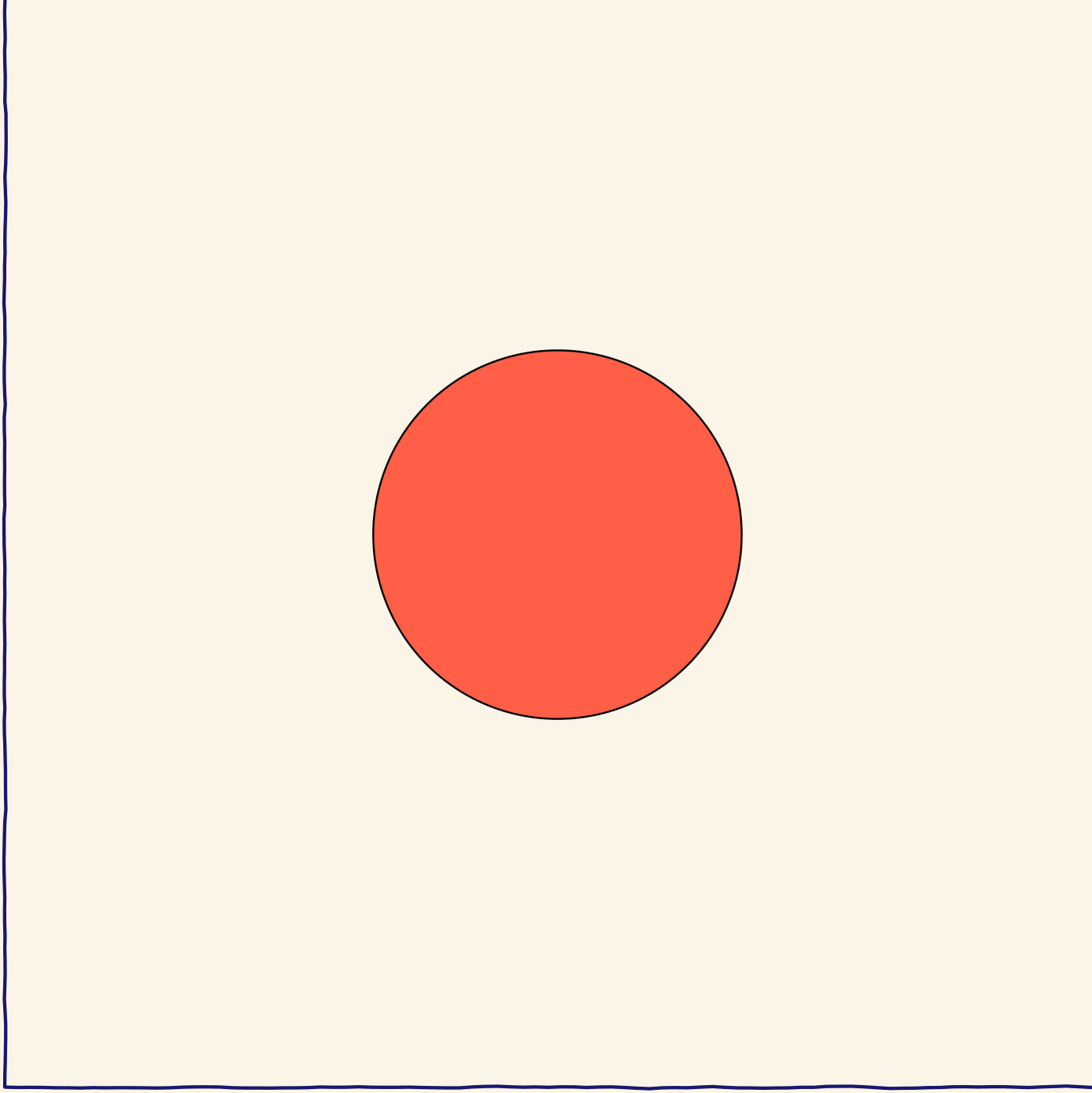
```
1 import funground as f
2
3 f.size(600, 600)
4 f.background("oldlace")
5 f.fill("tomato")
6 f.circle()
7 f.show()
```

The canvas: 600 wide, 600 high.

A colour is a word, in quotes.

No numbers. The circle goes in the middle.

Opens the window.

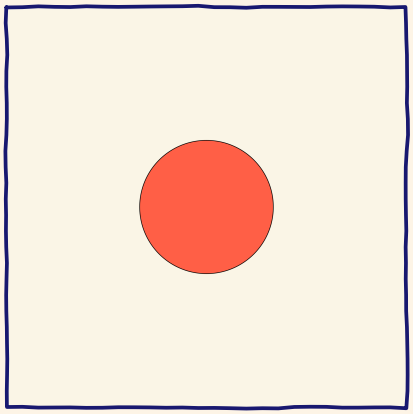


1.1 circle.py

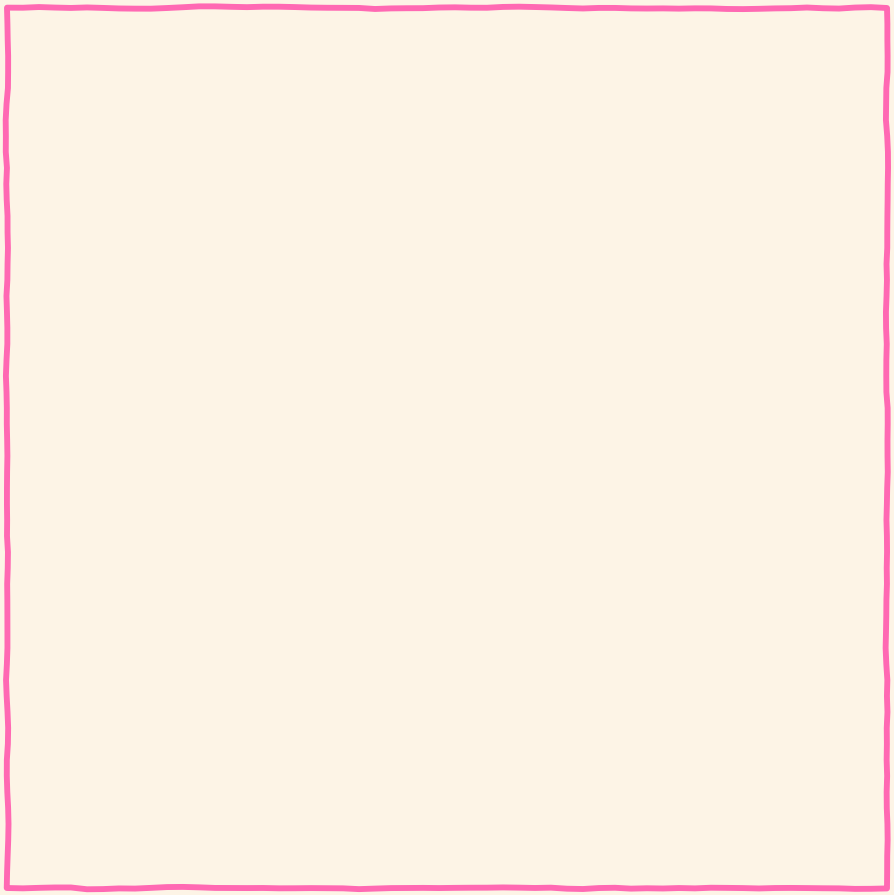
Add a square

Two new lines are in gold. Read them. What will the picture look like?

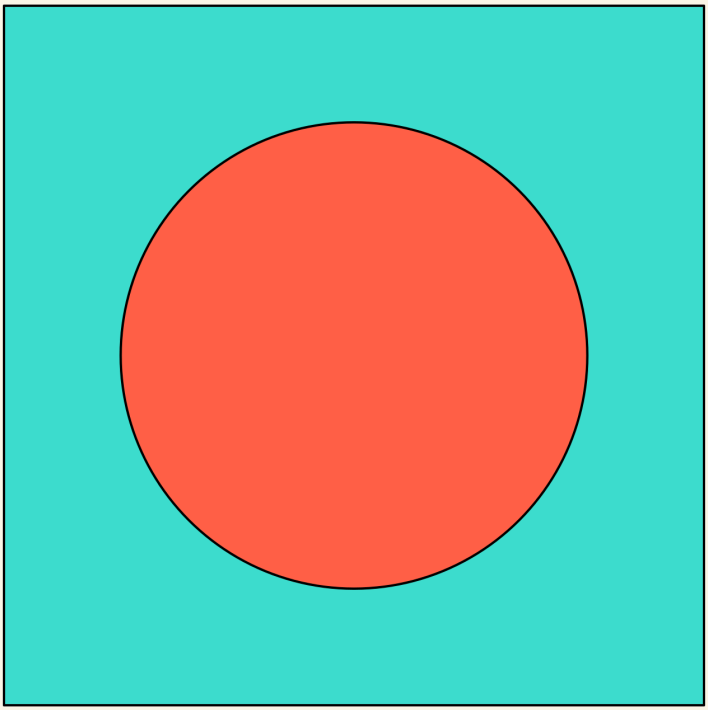
```
1  import funground as f
2
3  f.size(600, 600)
4  f.background("oldlace")
5  f.fill("turquoise")
6  f.square(size=300)
7  f.fill("tomato")
8  f.circle()
9  f.show()
```



what you know: 1.1 circle.py



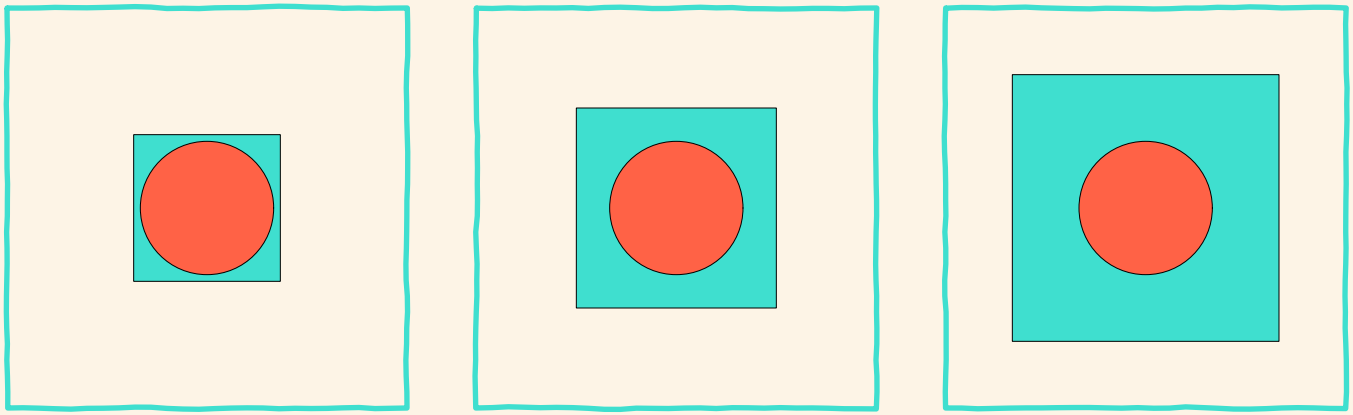
What will change? Draw it.



1.2 square.py

Change the size

The square is 300 wide. Here are three sizes, side by side. Only size= changes.



size=220 size=300 size=400

Try this
Change size=300 to 500. Then give the circle a size of its own.

Somewhere random

Say at="random" and the shape lands somewhere new. Each call picks its own place.

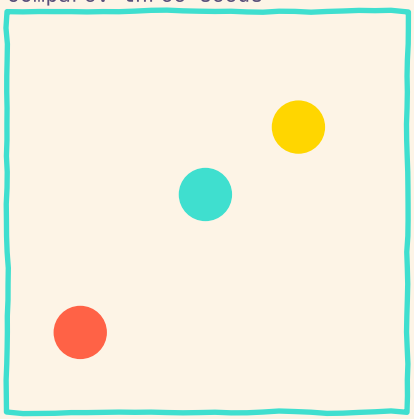
No outline: fills only.

at="random" picks a place. size=80 is the width.

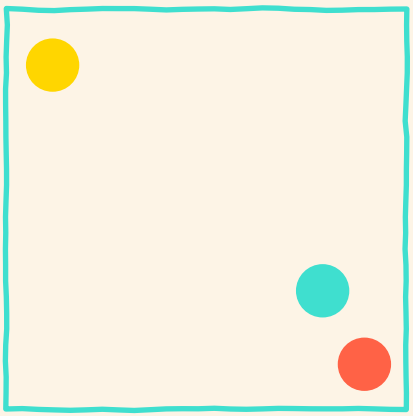
Same call. A new place.

```
1  import funground as f
2
3  f.size(600, 600)
4  f.background("oldlace")
5  f.no_stroke()
6  f.fill("tomato")
7  f.circle(at="random", size=80)
8  f.fill("gold")
9  f.circle(at="random", size=80)
10 f.fill("turquoise")
11 f.circle(at="random", size=80)
12 f.show()
```

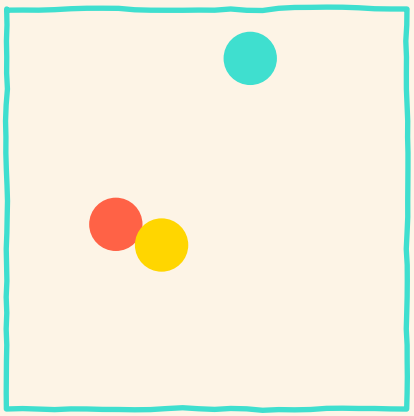
Compare: three seeds



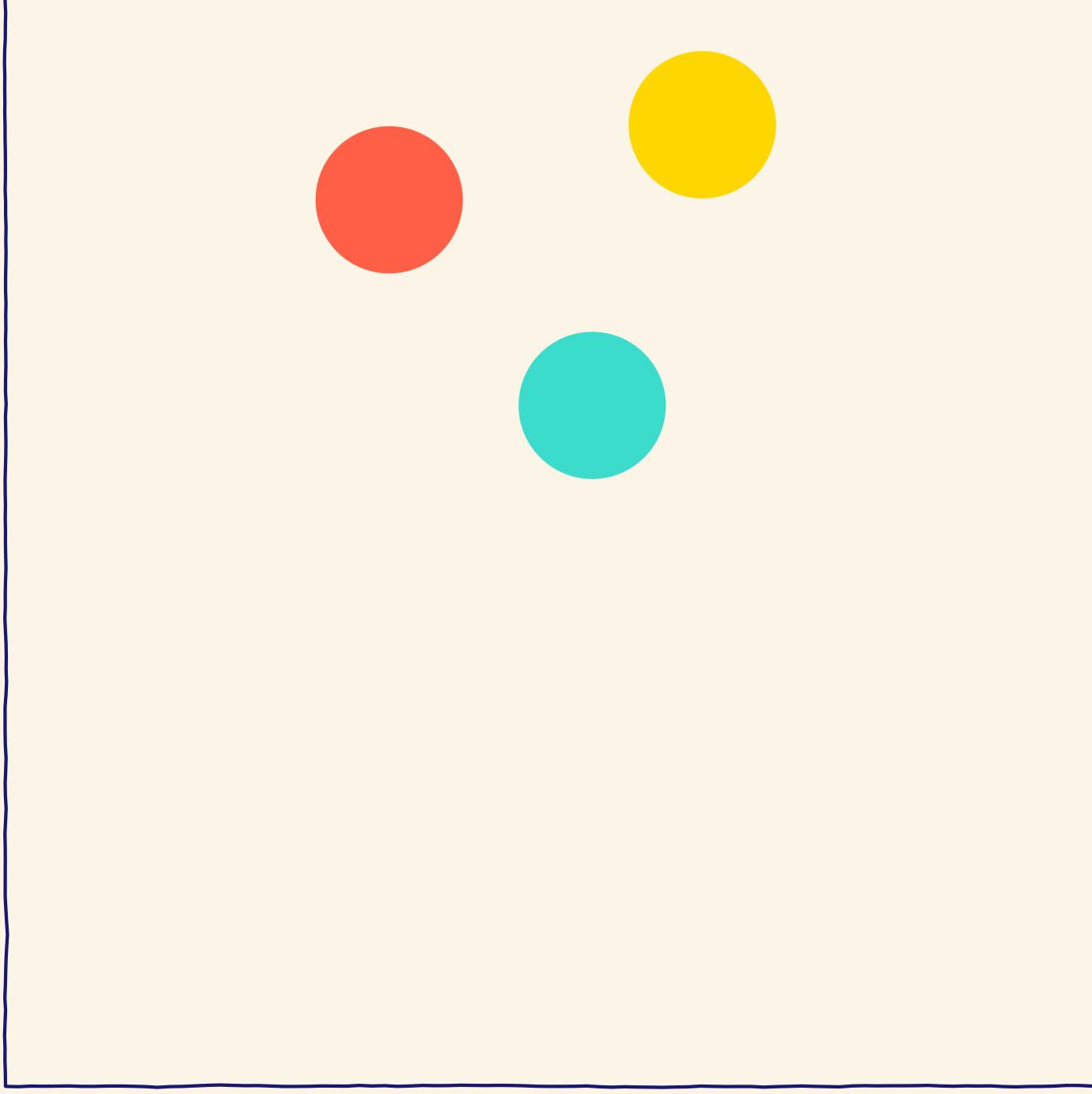
seed=1



seed=2



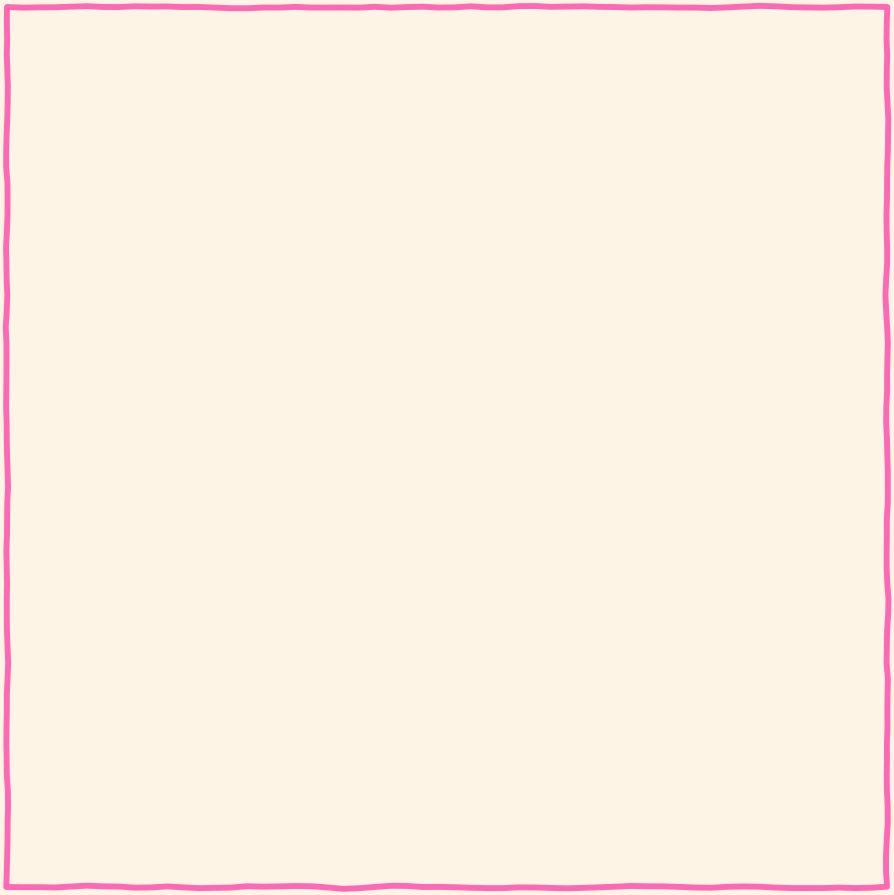
seed=3



1.3 random.py

Your page

Draw any shape. Change its colour, its size, and where it goes. Which change surprised you most?



Your canvas. Draw here, then try it in Play.

Colophon

Set in **Bricolage Grotesque**, Atkinson Hyperlegible Next and `JetBrains Mono`. All three are under the SIL Open Font License 1.1, which allows printing and embedding in a PDF.

The pictures are drawn by the sketches printed in this book. A program ran each one and placed its picture beside its code.

Made with funground. This chapter was laid out by a funground script: the grid, the type, the colours and the hand-drawn lines.

Paper is oldlace. Ink is midnightblue. The code in this book is CC0.